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Screening of Additives for Rapid Carbon Mineralization in Saline Aquifers

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Abstract

Atmospheric concentrations of carbon dioxide (CO₂) continue to increase at a high rate due to the burning of fossil fuels. Hence, Carbon Capture and Storage (CCS) are needed on a large scale and over time to cater this issue. Mineral carbonation differs among all CCS strategies about providing the most certain and permanent route of sequestration by transforming the CO₂ into stable solid carbonates; therefore, it is the most secure process. Nevertheless, one of the main shortcomings of natural mineralization is slow reaction rates, which are commonly several hundred and thousands of years at normal subsurface reservoir conditions. This paper explores a new, hastened format of CO₂ mineralization with the mixture of chemical motivators- Barium Hydroxide (Ba(OH)₂), Strontium Chloride (SrCl₂), and Glutamic Acid Diacetate (GLDA) into a synthetic prepared brine representing saline aquifer structures. Several laboratory experiments were run at a constant reservoir-pertinent temperature of 60 °C utilizing an artificial-brine and dolomite matrix. CO₂ gas was bubbled through the solutions with various additive concentrations; pH evolution and solid precipitation were tracked over 48 hours. The findings show that some combinations, especially 0.02 M Ba(OH)₂, 0.02 M SrCl₂, coupled with 0.1% GLDA had showed fast reactions in alkalinity and color carbonate formation, which was ascertained through XRD. The results challenge the traditional assumptions of the periods necessary in terms of the geological storage of CO₂ and the possibility of chemical augmentation to be subjected to mineralization in hours. The study, therefore, addresses a very large literature gap, providing a potentially scalable and productive pathway towards rapid CO₂ mineralization within carbonate aquifers, and with direct related implications to the application of large-scale CCS technologies.

Introduction

With the growing effects of climate change, it is now very important to lower the levels of carbon dioxide in the atmosphere. Increased use of Fossil fuels like Coal, oil, and Natural gas, which have been a very crucial source of energy since the Industrial Revolution, has become an evident result of this [1-3]. Total energy-related CO₂ emissions increased by 0.8% in 2024, hitting an all-time high of 37.8 Gt CO₂, which caused 2024 to become the first year where the temperature was more than 1.5 degrees Celsius [4]. The issue has led to growing and more significant attention from people across the globe on research on Carbon Capture and Storage (CCS). The methods used to trap CO₂ are structural trapping, residual trapping, solubility trapping, and mineral trapping [5-7]. Carbon mineralization refers to the natural or engineered process by which CO₂ reacts with metal oxides, primarily magnesium and calcium, to form solid carbonates such as magnesite (MgCO₃) or calcite (CaCO₃) [8]. Since CO₂ mineralization blocks the movement of stored carbon, it is considered the most reliable way to keep carbon from escaping [6]. Facilitating long-term storage, the process of carbon mineralization can be described as demonstrated in Figure 1.

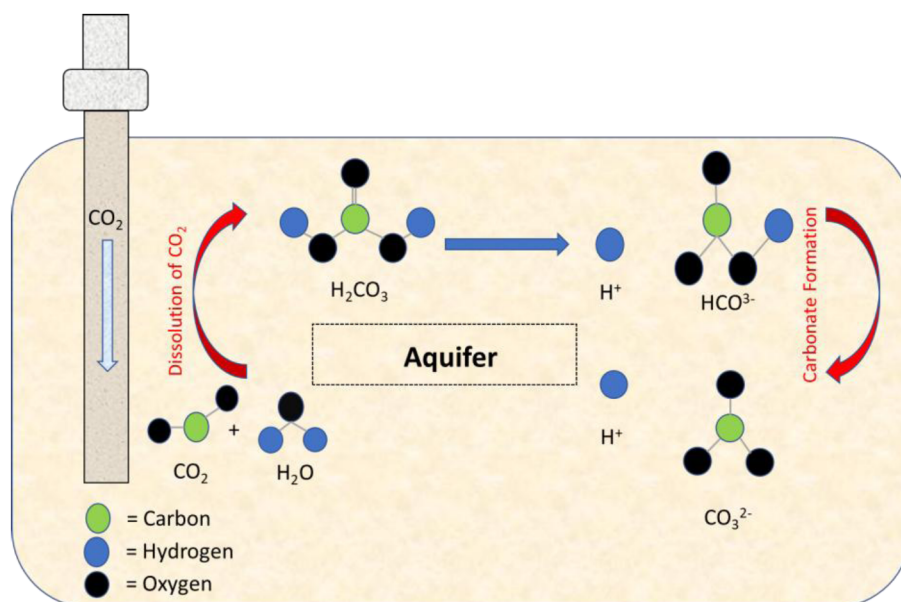


Figure 1—Schematic of Carbon Mineralization Process in saline aquifer

CO₂ initially dissolves in the water and forms bicarbonate and carbonate ions, and later dissolution of minerals produces rich solutions with the cations Ca²⁺, Mg²⁺, and Fe²⁺. Carbonates are finally produced through ion exchange in a high-pH solution. The efficiency of CO₂ mineralization often depends on the type of minerals present in the host material. Most discussions of CO₂ mineralization involve carbonate, mafic, and ultramafic rocks because their minerals react strongly [9]. Due to the presence of very few active minerals in quartz-rich sandstones and low injectivity, they are not suited for CO₂ storage.

Early research has shown that basalt and peridotite stand out as valuable rocks for mineralization because they have a high presence of magnesium and mineral cations [6,10]. When it comes to these rocks, the fastest reactions are found in ultramafic igneous rocks, and the slowest in common sedimentary rocks [11]. As it could take thousands of years for carbon mineralization in reservoir rocks, this form of CO₂ storage is mostly ignored in the literature, except for a few studies that focus solely on the modifications in the rock's properties caused by injecting CO₂ [12]. In the early years, Sass et al. (2001) conducted experiments on mineral rocks being exposed to brine for seven days under 5.44 MPa and 110 °C. They found that dissolving dolomite in brine produced more calcium and magnesium [13]. Moreover, the authors said that the fall in aqueous calcium and sulfate levels resulted from the formation of anhydrite. However, there has been a

recent study by [White et al. 2020](#) in basaltic rock, which took them only two years to complete mineralization [14]. These instances depict a clear literature gap for the timeline to form CO₂ mineralization and need to innovate new methods to enhance Mineralization (reduce time duration) of CO₂.

The process of subsurface CO₂ mineralization is complex, since it depends on temperature, pressure, and salinity. Higher pressure supports the mineralization process, while higher temperature or salinity work against it. CO₂ might "precipitate out" if a saline layer is present, therefore reducing its amount in the water, while increasing ionic strength also changes the values of carbonate equilibrium constants, causing dissolution to slow down and a reduction in volume. As the CO₂ dissolves in the formation water, the pH decreases up to typically 3.0–5.0 [15]. This acidity at the beginning allows the acid to dissolve minerals and unleash Ca²⁺/Mg²⁺, but after some time, it gets neutralized when carbonates begin to form. When the pH level is over 8–9, dissolution stops, and pores can get blocked. Equation 1 and 2 shows ionic interaction to form carbonates under high pH conditions.



The product of CO₂ dissolution reacting with carbonate minerals in the rock adds another level of complications, which means that the dissolution and secondary mineral precipitation caused by the injection of CO₂ have a major impact on the porosity and permeability of the reservoir environment, as well as impact the cap rock integrity [16]

Furthermore, it has been observed that the petrophysical properties of rock are also impacted by the interaction of CO₂-rich fluid with the rock. Wigand et al. found that entrapment of CO₂ in deep saline aquifers is mainly due to capillary and dissolution forces and is little influenced by mineral trapping [17], which was later confirmed by Raza et al. [18]. However, Marbler et al. reported that under the interaction of CO₂-brine with carbonate and sandstone rock, the friction angle and pore space get altered while the rock strength becomes weaker as a result of carbonate mineral formation on the exposed surfaces [19]. Rathnaweera et al. explored something similar, where they discovered that changes in the pore structure in the used rock sample due to CO₂ exposure led to a better permeability [20]. In contrast to this, [Hangx et al. \(2013\)](#) found in their study that CO₂ did not cause any changes in the rock mechanics of the sample [21]. Thus, highlighting the need for more targeted research to clarify these effects.

In essence, successful carbon mineralization in the subsurface depends on good CO₂ dissolution, speedy reaction with cations, and avoidance of clogging at any stage of mineralization. Strategies for resolving the issue are chemical additions (acids, organic ligands), better CO₂ injection (brine with CO₂, pressure variations), and choosing suitable locations (with plenty of reactive minerals and the right temperature). Recent research has also investigated chemical additives that can stimulate the process, such as SrCl₂ in anhydrite, for rapid mineralization [22] (or that can tip the carbonate equilibrium towards solid formation (e.g., alkalis) to improve it). Much of the experimental work has concentrated on single-component systems or ultramafic rocks and less on carbonate aquifer systems like those hosted in dolomite brines. The objective of the current study is to fill that gap by assessing the synergistic effect of Ba(OH)₂, SrCl₂, and GLDA on CO₂ mineralization rate under laboratory conditions that model a carbonate saline aquifer. These goals are twofold:

1. To determine the additive compositions to optimize pH buffering, maximize cation availability, and enhance quick carbonate formation.
2. To show that it is possible to achieve visible mineralization in a matter of hours rather than years.

GLDA (Glutamic Acid Diacetate) has been employed as a biodegradable chelating agent to regulate metal ion availability by the process of leaching from the rock surface [23], while alkaline compounds such

as $\text{Ba}(\text{OH})_2$ can rapidly elevate pH to facilitate carbonate precipitation and are more feasible because of having high reactivity, low-cost recyclability, and better selectivity[24]. This study contributes by providing an understanding of the potential massive acceleration of CO_2 mineralization by using specific chemical formulations, as seen through a series of screening tests in which synthetic brine, powdered dolomite, and direct injection of CO_2 were conducted at 60 °C. It also leads to improved understanding of the relative roles of brine chemistry, additive concentration, and precipitation dynamics here, and gives a means of hypothetical future applications at high pressure and scale-up in real applications of CCS.

Methodology

The main purpose of this research is to examine how $\text{Ba}(\text{OH})_2$, SrCl_2 , and GLDA affect the process of CO_2 mineralization in a controlled environment. Additionally, this research will examine the reaction together with the carbonate precipitation efficiency and brine environment pH stability as a result of the additive's exposure. However, the study identifies ideal experimental parameters for enhanced CO_2 storage, along with mineralization, in a laboratory setting. The flow chart in Figure 2 below describes the procedure for this.

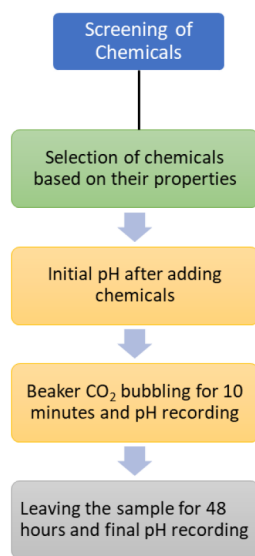


Figure 2—The steps followed for the screening of chemicals

Experimental Setup

A 125 mL flask was used to test a synthetic seawater brine solution, which reflects the chemical profile of natural formation brine as tabulated in Table 1. Sodium bicarbonate (NaHCO_3), along with sodium sulfate (Na_2SO_4) and sodium chloride (NaCl), together with calcium chloride ($\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$) and magnesium chloride ($\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) will compose the brine solution to replicate formation brine conditions in natural settings. A combination of $\text{Ba}(\text{OH})_2$, SrCl_2 , and GLDA at different concentrations was added to the brine solution before it was exposed to CO_2 to improve mineral formation. The experimental setup can be understood by the schematic Figure 3 below.

Table 1—Synthetic brine compositions

Species	Mass Used, g	Concentration, g/L
NaCl	205.86	41.172
MgCl ₂ ·6H ₂ O	88.22	8.266
CaCl ₂ ·2H ₂ O	11.935	1.802
Na ₂ SO ₄	31.695	6.339
NaHCO ₃	0.825	0.165
Total Dissolved Solids		57.745

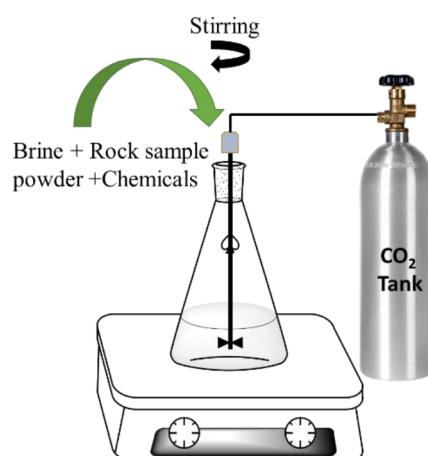


Figure 3—Schematic of the static setup used for treatment at 60°C

As a starting point for the experiment, 100 mL of brine was added to the flask, followed by adding the additives at selected compositions, followed by the addition of 10 g of powdered dolomite ($\text{CaMg}(\text{CO}_3)_2$) for simulating natural mineral surfaces. The flask containing the solution was maintained at a constant 60°C temperature using a hot plate, after which a magnetic stirrer maintained solution uniformity throughout the test. The initial pH, after adding chemical additives, was recorded for that concentration. Now, CO_2 gas was passed through the heated beaker by bubbling for 10 minutes using direct bubbling to speed up both dissolution and carbonate precipitation, followed by pH measurement at each step. The examination of the mineralization process was done at selected points in time to determine its advancement. This study measures reaction together with carbonate precipitation while recording pH modification data at predefined time points. The knowledge about mineralization efficiency will be determined by measuring the final precipitate white deposits formed and confirmed visually. The carried-out investigation delivers preliminary findings about reaction timing and effectiveness, which can be used to determine elevated-pressure system configurations. The whole test was repeated at different compositions of all three chemicals to get the required pH levels.

Results and Discussion

The section will give a critical appraisal of the experimental data in 11 test cases with different ratios of $\text{Ba}(\text{OH})_2$, SrCl_2 , and GLDA. Modification of pH, appearance of precipitation, increase of turbidity, and efficiency of carbonate formation are the basis of analysis.

pH behavior and additives influence

The values of pH development through the 11 test cases (Table 2) exhibit clear tendencies of buffering performance and efficiency of precipitation. The base (Case 1) containing no additives reached a final pH

of 6.0, which is characteristic of the brine-CO₂ interactions, and in agreement with [Wigand et al. \(2008\)](#) results [17]. Cases with Ba(OH)₂ (e.g., Case 3 and Case 6) already had much higher initial pH (up to 9.58) and had final pH stability that was better, suggesting better conditions regarding carbonate formation. This observation compares to those of [Zevenhoven et al. \(2014\)](#), who stated that carbonate saturation states are achieved more easily in the presence of alkaline additives [25]. Furthermore, application of SrCl₂ (Case 2) only slightly, whereas when combined with other cases (Cases 5-11), it seemed to facilitate other possible precipitation routes, probably the formation of SrCO₃. Incidentally, GLDA acted in two ways. At low concentrations, it buffered pH without affecting cation activity (Case 4), at 0.2% (Cases 7, 10, 11), it suppressed initial pH, perhaps by complexing with Ca²⁺ and Mg²⁺. This confirms earlier observations of [Othman et al. \(2021\)](#) regarding pH and GLDA concentration[26].

Table 2—Screening behavior of different chemicals

Additives	Strontium Chloride (SrCl ₂)	Barium Hydroxide (Ba(OH) ₂)	Glutamic Acid Diacetate (GLDA)	pH		
				Initially, after adding chemicals	After CO ₂ bubbling	Final after 48 hours
Case 1	-	-	-	7.55	5.77	6
Case 2	0.01 M	-	-	7.43	5.6	6.07
Case 3	-	0.01 M	-	9.15	5.68	6.5
Case 4	-	-	0.1 %	5.57	5.36	6.42
Case 5	0.01 M	0.01 M	0.1 %	8.76	5.54	6.23
Case 6	0.01 M	0.02 M	0.1 %	9.58	6.17	6.43
Case 7	0.01 M	0.01 M	0.2 %	5	5.43	5.96
Case 8	0.02 M	0.01 M	0.1 %	7.2	5.47	6.45
Case 9	0.02 M	0.02 M	0.1 %	8.98	5.54	6.42
Case 10	0.01 M	0.02 M	0.2 %	6.25	5.9	6.72
Case 11	0.02 M	0.01 M	0.2 %	5.56	5.76	6.08

Visual observation of precipitation and turbidity

Precipitation was visually confirmed in all cases of Ba(OH)₂, and thick white layers of BaCO₃ and CaCO₃ were visualized and confirmed by XRD, which was also the case in Carbfix pilot tests in Iceland. The turbidity was the highest in Case 9, and this affirmed that optimized mixtures (0.2 M of Ba(OH)₂ + 0.2 M of SrCl₂ + 0.1% of GLDA solution) promoted fast and efficient precipitation of carbonates [27]. The majority of the available researchers assert that the process of mineralization in the reservoir rock should take years to be concluded [28]. Nevertheless, we do report carbonate precipitation on the hour-timescales with the kinetically governed surface area or additive concentration increase as documented by [Schott et al. \(2012\)](#) [29]. This affirms the practicality of synthetic improvement in the procedure of mineralization.

Mineralogical Analysis

In XRD analysis of the solids of Case 1 and Case 9, respectively, dolomite is the predominant mineral phase of both specimens. In Case 9, the dolomite peaks are visible sharper and stronger, which means higher crystallinity and higher complete reaction. Case 1, however, depicts a combination of dolomite with some minor heights that could belong to other carbonate minerals. This indicates that the reaction of Case 1 was not as complete as that of Case 9. All in all, it can be observed through XRD patterns ([Figure 3](#)) that the case of 9 experienced a more efficient carbonation process, which can be attributed to the possible better cation

availability. The calculated dolomite fraction rose to about 92.3% (Case 9) compared to about 85.5% (Case 1) in the formation of dolomite, favoring the positivity in the mineral trapping capacity of CO₂ in dolomite.

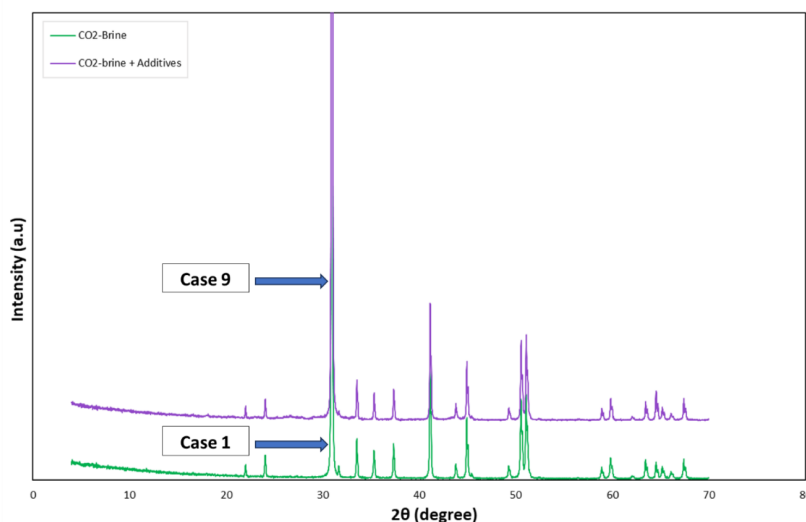


Figure 4—XRD analysis of Case 1 and Case 9

Target pH profile and kinetic considerations

The progression through targeted pH stages is crucial for efficient CO₂ mineralization. A pre-injection high pH (8-10) favors the formation of carbonate ions and increases cation solvation, which leads to good conditions of early nucleation [24]. The acidity pH value changes to 4-6 upon injection of CO₂, where the minerals are dissolved and the important ions like Ca²⁺, Mg²⁺, and Ba²⁺. Such an acidic window foregoes the enrichment of the brine with active species, speeding up geochemical reactions [15]. As the carbonation continues, pH is maintained in line between 6 to 10, allowing precipitation to occur, in the stable state over a long period as carbonates such as calcite and magnesite [30]. This final pH range avoids any over-saturation, pore plugging, and keeps reservoir injectivity intact. The sequence of pH management increases the kinetics of mineralization, safe storage over the long term, and the large-scale permanent CO₂ immobilization in subsurface salty aquifers. The pH values were targeted according to Figure 3 below for boosting the mineralization process. However, the final pH after the mineralization process depends upon the time provided to the system to form stable carbonates and consume CO₂.

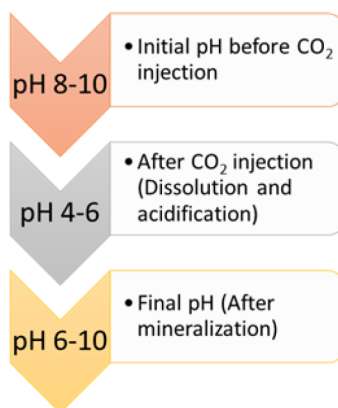


Figure 5—pH evolution during the CO₂ mineralization process, highlighting three key stages.

Conclusion

The research establishes the fact that the process of CO₂ mineralization in carbonate aquifers can be enhanced by utilizing specially selected chemical additives. In particular, Ba(OH)₂, SrCl₂, and GLDA mixtures at elevated aquifer temperatures (60 °C) promoted rapid pH buffering and extensive carbonate precipitation in hours. The findings not only support but also extend the current literature, indicating the acceleration of reactions by engineering brine chemistry. In contrast to previous work, which either needed high pressures or rare mineral formation, this study demonstrates that common chemicals and easy-to-construct arrangements can be used to accomplish the same.

Beyond mineralization, this study preconditions the assessment of petrophysical property alterations through SEM and core flooding in the future, in line with the procedures undertaken by Rathnaweera et al. (2016) and Hangx et al. (2013) [20,21]. The findings have potential for being used in a real sense in carbonates-rich reservoirs of the Middle East, where trapping of CO₂ is rapid in the implementation of large-scale CCS. Moreover, when applied to subsurface mineralization, feedback of rock matrix dissolution and carbonate mineral precipitation on petrophysical flow properties of the Carbonate aquifer requires further study to improve understanding. There are also further uncertainties over the scalability of mineralization operations because mineral trapping has been mainly proven at laboratory scale, with only a handful of pilot-scale subsurface injection projects [31]. In light of the need to find a solution in the management of carbon and with the pressure rising on finding a feasible solution, the study provides a feasible competent solution to the improvement in the secure sequestration of CO₂ in carbon-bearing formations which is especially in the geologic context of the Middle East and other carbon capture and storage target areas around the world.

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References

1. M.D. Aminu, S.A. Nabavi, C.A. Rochelle, V. Manovic, A review of developments in carbon dioxide storage, *Appl. Energy* **208** (2017) 1389–1419. <https://doi.org/10.1016/J.APENERGY.2017.09.015>.
2. S. Khandoozi, R. Hazlett, M. Fustic, A critical review of CO₂ mineral trapping in sedimentary reservoirs – from theory to application: Pertinent parameters, acceleration methods and evaluation workflow, *Earth-Science Rev.* **244** (2023) 104515. <https://doi.org/10.1016/j.earscirev.2023.104515>.
3. A. Raza, R. Gholami, R. Rezaee, V. Rasouli, M. Rabiei, Significant aspects of carbon capture and storage – A review, *Petroleum* **5** (2019) 335–340. <https://doi.org/10.1016/j.petlm.2018.12.007>.
4. *Global Energy Review 2025*, n.d. www.iea.org.
5. O. Massarweh, A.S. Abushaikh, CO₂ sequestration in subsurface geological formations: A review of trapping mechanisms and monitoring techniques, *Earth-Science Rev.* **253** (2024) 104793. <https://doi.org/10.1016/j.earscirev.2024.104793>.
6. A. Raza, G. Glatz, R. Gholami, M. Mahmoud, S. Alafnan, Carbon mineralization and geological storage of CO₂ in basalt: Mechanisms and technical challenges, *Earth-Science Rev.* **229** (2022) 104036. <https://doi.org/10.1016/j.earscirev.2022.104036>.
7. F. Khan, M. Mahmoud, A. Raza, S. Patil, M.J. AlTammar, T. Al Shafloot, Application of Cold CO₂ Injection in High Skin Factor Depleted Reservoirs: A Review, in: *GOTECH*, SPE, 2024. <https://doi.org/10.2118/219351-MS>.

8. A. Raza, R. Rezaee, R. Gholami, C.H. Bing, R. Nagarajan, M.A. Hamid, A screening criterion for selection of suitable CO₂ storage sites, *J. Nat. Gas Sci. Eng.* **28** (2016) 317–327. <https://doi.org/10.1016/j.jngse.2015.11.053>.
9. H. Wang, A. Nguyen, T. Wang, R. Okuno, V.F.H. Serra, P.T.P. Aum, Enhanced CO₂ Mineralization in Stromatolite and Magnesium and Calcium Silicates Using Formate Ligands, in: SPE Int. Conf. Oilf. Chem., SPE, 2025. <https://doi.org/10.2118/224315-MS>.
10. Y. Pan, B. Ji, B. Nojabaei, R. Pandey, W. Zhang, Investigating CO₂ mineralization of fine mafic minerals: Insights into reaction characteristics and physicochemical attributes, *J. Environ. Chem. Eng.* **13** (2025) 116827. <https://doi.org/10.1016/j.jece.2025.116827>.
11. P. Kelemen, S.M. Benson, H. Pilorgé, P. Psarras, J. Wilcox, An Overview of the Status and Challenges of CO₂ Storage in Minerals and Geological Formations, *Front. Clim.* **1** (2019). <https://doi.org/10.3389/fclim.2019.00009>.
12. L. Karabayanova, M. Hajiyevev, M. Isabel Hernandez Madero, B.E. Sutherland, J.B. Harris, I. Qureshi, B. Hascakir, Accelerated Carbon Capture and Storage via Mineral Carbonation: A Laboratory-Scale Investigation, in: SPE Annu. Tech. Conf. Exhib., SPE, 2024. <https://doi.org/10.2118/220815-MS>.
13. B. Sass, N. Gupta, J. Ickes, M. Engelhard, D. Baer, P. Bergman, C. Byrer, E. Boyle, Interaction of Rock Minerals with Carbon Dioxide and Brine: A Hydrothermal Investigation, *J. Energy Environ. Res.* **2**(1) (2001).
14. S.K. White, F.A. Spane, H.T. Schaef, Q.R.S. Miller, M.D. White, J.A. Horner, B.P. McGrail, Quantification of CO₂ Mineralization at the Wallula Basalt Pilot Project, *Environ. Sci. Technol.* **54** (2020) 14609–14616. <https://doi.org/10.1021/acs.est.0c05142>.
15. K. Kim, D. Kim, Y. Na, Y. Song, J. Wang, A review of carbon mineralization mechanism during geological CO₂ storage, *Heliyon* **9** (2023) e23135. <https://doi.org/10.1016/j.heliyon.2023.e23135>.
16. X. Li, R. Xu, P. Jiang, Modelling and interpretation of a temperature drop in the wellbore in Ordos CO₂ storage project, China, *Energy Procedia* **63** (2014) 6413–6419. <https://doi.org/10.1016/j.egypro.2014.11.676>.
17. M. Wigand, J.W. Carey, H. Schütt, E. Spangenberg, J. Erzinger, Geochemical effects of CO₂ sequestration in sandstones under simulated in situ conditions of deep saline aquifers, *Appl. Geochemistry* **23** (2008) 2735–2745. <https://doi.org/10.1016/j.apgeochem.2008.06.006>.
18. A. Raza, R. Rezaee, C.H. Bing, R. Gholami, M.A. Hamid, R. Nagarajan, Carbon dioxide storage in subsurface geologic medium: A review on capillary trapping mechanism, *Egypt. J. Pet.* **25** (2016) 367–373. <https://doi.org/10.1016/j.ejpe.2015.08.002>.
19. H. Marbler, K.P. Erickson, M. Schmidt, C. Lempp, H. Pöllmann, Geomechanical and geochemical effects on sandstones caused by the reaction with supercritical CO₂: an experimental approach to in situ conditions in deep geological reservoirs, *Environ. Earth Sci.* **69** (2013) 1981–1998. <https://doi.org/10.1007/s12665-012-2033-0>.
20. T.D. Rathnaweera, P.G. Ranjith, M.S.A. Perera, Experimental investigation of geochemical and mineralogical effects of CO₂ sequestration on flow characteristics of reservoir rock in deep saline aquifers, *Sci. Rep.* **6** (2016) 19362. <https://doi.org/10.1038/srep19362>.
21. S. Hangx, A. van der Linden, F. Marcelis, A. Bauer, The effect of CO₂ on the mechanical properties of the Captain Sandstone: Geological storage of CO₂ at the Goldeneye field (UK), *Int. J. Greenh. Gas Control* **19** (2013) 609–619. <https://doi.org/10.1016/j.ijggc.2012.12.016>.
22. A. Isah, M. Mahmoud, A. Raza, M. Murtaza, M. Arif, M.S. Kamal, CO₂-brine interactions in anhydrite-rich rock: Implications for carbon mineralization and geo-storage, *Int. J. Greenh. Gas Control* **137** (2024) 104202. <https://doi.org/https://doi.org/10.1016/j.ijggc.2024.104202>.

23. J. Al-Darweesh, M. Saleh Aljawad, M. Shahzad Kamal, M. Mahmoud, Z. Al-Yousef, The impact of chelating agent pH on the stability and viscosity of CO₂ foam under harsh reservoir conditions, *J. Mol. Liq.* **412** (2024) 125847. <https://doi.org/https://doi.org/10.1016/j.molliq.2024.125847>.
24. J. Wang, L. Li, G. Qu, M. Cheng, Y. Ren, P. Lu, T. Zhang, Y. Wang, F. Wu, Advanced materials for CO₂ capture and storage: A comprehensive review of current progress and future prospects, *J. Environ. Chem. Eng.* **13** (2025) 115881. <https://doi.org/https://doi.org/10.1016/j.jece.2025.115881>.
25. R. Zevenhoven, J. Kohlmann, K. Phaal, *Carbon Dioxide Sequestration by Mineral Carbonation: Literature Review Update 2005–2007*, Åbo Akademi University, Finland, 2008. <https://remineralyze.org/wp-content/uploads/2015/10/LITR1.pdf>.
26. A. Othman, M.S. Aljawad, M. Mahmoud, M.S. Kamal, S. Patil, M. Bataweel, Chelating Agents Usage in Optimization of Fracturing Fluid Rheology Prepared from Seawater, *Polymers (Basel)*. **13** (2021) 2111. <https://doi.org/10.3390/polym13132111>.
27. T.M.P. Ratouis, S.Ó. Snæbjörnsdóttir, M.J. Voigt, B. Sigfússon, G. Gunnarsson, E.S. Aradóttir, V. Hjörleifsdóttir, Carbfix 2: A transport model of long-term CO₂ and H₂S injection into basaltic rocks at Hellisheidi, SW-Iceland, *Int. J. Greenh. Gas Control* **114** (2022) 103586. <https://doi.org/10.1016/j.ijggc.2022.103586>.
28. L.M. Hamm, I.C. Bourg, A.F. Wallace, B. Rotenberg, Molecular Simulation of CO₂- and CO₃-Brine-Mineral Systems, *Rev. Mineral. Geochemistry* **77** (2013) 189–228. <https://doi.org/10.2138/rmg.2013.77.6>.
29. J. Schott, O.S. Pokrovsky, E.H. Oelkers, The Link Between Mineral Dissolution/Precipitation Kinetics and Solution Chemistry, *Rev. Mineral. Geochemistry* **70** (2009) 207–258. <https://doi.org/10.2138/rmg.2009.70.6>.
30. J.M. Matter, M. Stute, S.Ó. Snæbjörnsdóttir, E.H. Oelkers, S.R. Gislason, E.S. Aradóttir, B. Sigfússon, I. Gunnarsson, H. Sigurdardóttir, E. Gunnlaugsson, G. Axelsson, H.A. Alfredsson, D. Wolff-Boenisch, K. Mesfin, D.F. de la R. Taya, J. Hall, K. Dideriksen, W.S. Broecker, Rapid carbon mineralization for permanent disposal of anthropogenic carbon dioxide emissions, *Science (80-.)*. **352** (2016) 1312–1314. <https://doi.org/10.1126/science.aad8132>.
31. S.Ó. Snæbjörnsdóttir, B. Sigfússon, C. Marieni, D. Goldberg, S.R. Gislason, E.H. Oelkers, Mineral storage of CO₂, *Nat. Rev. Earth Environ.* **1** (2020) 90–102.